Wonguk Lee

Business and Technology Management
Korea Advanced Institute of Science and Technology
Ph.D. Student, Management Information Systems
Advisor: Prof. Chul Ho Lee

E-mail: lwg5253@kaist.ac.kr
Web: lwg52534-dev.github.io

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Baekseok University, Fall 2026 — Lead Course Instructor
TalentSeeker, February 2024 - present — Chief Data Officer

EDUCATION

Ph.D. Student, Business and Technology Management (MIS), KAIST
M.S., Business and Technology Management (MIS), KAIST
B.B.A., Business Administration, Handong Global University
B.A., Law (Double Major), Handong Global University

RESEARCH INTERESTS

Conversational AI and mental healthcare
Mobility and geographic access to essential services
Digital platforms and AI governance
Technological innovation and knowledge flows

PUBLICATIONS AND CONFERENCE PROCEEDINGS

Kim, Junehwan, Wonguk Lee, and Euiseok Kim*, forthcoming*, "Sectoral Heterogeneity in the Digital Transformation-Financial Performance Nexus: Evidence from Korean Manufacturing Firms," accepted in ***Asian Journal of Technology Innovation***.

Lee, Wonguk, Donggyu Min, Antino Kim, and Chul Ho Lee (2026), "Thought-Bubbles in Therapy: The Effect of Conversational AI's Transparency in Mental Health Contexts," ***ICIS 2026 Proceedings***, Healthcare and Wellbeing, Paper 19. Proceedings available; presentation forthcoming, December 2026. Proceedings

Shenkoya, Temitayo, Wonguk Lee, and Euiseok Kim (2024), "Transformation of Business Models in Metaverse Era: Through a Systematic Literature Review and Bibliometric Analysis," ***Journal of Information Technology Services***, 23(6), 1-17. DOI: 10.9716/KITS.2024.23.6.001

Lee, Wonguk and Euiseok Kim (2024), "A Study on the Flow of Technological Knowledge in Korean Industries: Focusing on Inter-industry Patent Citation Analysis Based on Pavitt's Taxonomy," ***Journal of Korea Technology Innovation Society***, 27(3), 431-454. DOI: 10.35978/jktis.2024.6.27.3.431

Lee, Wonguk and Heetae Yang (2022), "A Study on the Intention to use Personal Mobility Services: Focused on the SOR (Stimulus-Organism-Response) Model," ***Information Systems Review***, 24(2), 67-88. DOI: 10.14329/isr.2022.24.2.067

WORKING PAPERS

Lee, Wonguk and Euiseok Kim, "Formation and Dissipation of the Hypernetwork Effect: Transitional Complementarity between Learning and Participation in AI-Driven Networks," Under revision · Second round.

Kim, Kyeong Hun, Wonguk Lee, and Byung Hwa Lim, "Primary Token Issuance and Liquidity-Segmented Returns in Crypto Markets," Under revision · Minor revision.

Yang, Heetae and Wonguk Lee, "Understanding corporate digital entrepreneurship in financial institutions: The roles of AI adoption and entrepreneurial orientation," Submitted.

Lee, Wonguk and Heetae Yang, "Understanding Personal Mobility Device Adoption Through the S-O-R Framework: A Multi-Group Analysis of Users and Non-users," Under review.

Kim, Kyusung, Wonguk Lee, Haeun Jang, and Hyungu Jang, "The Effects of Public Bike-Sharing (Tashu) and Public Transit Integration on Local Festivals and Consumer Spending," Under revision · Excellent Paper Award, Korea Society of IT Services.

Lee, Wonguk and Euiseok Kim, "The Role of NFTs in the Digital World: Catalyzing Innovation in the Creative Industry through a Patent-Inspired Approach," Under review.

WORK IN PROGRESS

Lee, Wonguk and Euiseok Kim, "Breadth to Enter, Relatedness to Stay: Technology Portfolios and the Limits of Firms' Green Diversification," Work in progress.

Lee, Wonguk, Donggyu Min, Antino Kim, and Chul Ho Lee, "Thought-Bubbles in Therapy: The Effect of Conversational AI's Transparency in Mental Health Contexts," Work in progress; related ICIS 2026 proceedings version listed above.

Lee, Wonguk, Gunsu Kim, and Chulho Lee, "Protecting the Investor, Punishing the Retailer: The Perverse Effects of Banning AI Autonomous Investment," Work in progress · Accepted for presentation at WITS; presentation forthcoming.

Lee, Wonguk and Chul Ho Lee, "Funded but Not Followed? Donor Relationships After Sponsor-Led Funding Completion," Work in progress.

Lee, Wonguk and Chul Ho Lee, "AI Summary Design in Consumer Complaint Platforms," Work in progress · Working title.

Books and Chapters

Lee, W. (2026), "Generative AI as a Creative Collaborator." In *AI Transformation Era: Innovation and Challenges of Korean Industries Highlighted by KAIST*. Yulgok Publishing.

AI Era: Structure and Sustainability of the Korean Workforce Ecosystem. Co-author. Selected as an Excellent Academic Book by the National Academy of Sciences, Republic of Korea (2025).

HONORS AND AWARDS

Best Paper Award, BTM Seminar, KAIST. Thought Bubbles in Therapy: The Effect of Conversational AI's Transparency in Stigmatized Mental Health Contexts

Excellent Paper Award. Korea Society of IT Services | Public bike-sharing (Tashu) and public-transit integration study.

Outstanding Research Idea Award. BTM Seminar, KAIST | When the AI Bot Advisor Leaves: Unpacking Investor Shifts and Loan Imbalances in P2P Lending

Excellent Academic Book Selection (National Academy of Sciences, Republic of Korea). AI Era: Structure and Sustainability of the Korean Workforce Ecosystem (Co-author)

UPCOMING CONFERENCE PRESENTATIONS

International Conference on Information Systems (ICIS 2026). Thought-Bubbles in Therapy: presentation forthcoming, December 2026. Conference: 13-16 December, Lisbon, Portugal. Proceedings available online.

CES 2027. Scheduled to attend as a C-level speaker in my role as Chief Data Officer at TalentSeeker.

Workshop on Information Technology and Systems. Protecting the Investor, Punishing the Retailer: accepted for presentation at WITS; presentation forthcoming.

TEACHING EXPERIENCE

Lead Course Instructor (Baekseok University) Fall 2026
Data Science (English-medium)
English-medium course design and teaching in Python, pandas, data preparation, visualization, statistics, prediction, and responsible generative AI use.

PROFESSIONAL EXPERIENCE

Chief Data Officer, TalentSeeker. February 2024-present. Full-time C-level role. Lead data ontologies, multi-agent systems, data-pipeline architecture, data engineering, and LLMOps for AI-enabled recruitment.

PROJECT EXPERIENCE

Onto-LLM-Based Global Outbound Recruitment Service Development. TIPS R&D Program, Ministry of SMEs and Startups (MSS) | Aug 2025 - Jul 2027 | Ongoing. Contribute to ontology-based recruitment AI, Graph-RAG / Graph-of-Thought reasoning, and agent prototype research.

Open-Web-Based Automated Resume Collection and AI Recruitment Support System. Startup Growth Technology Development Program, TIPA / MSS | Aug 2024 - Jul 2025 | Completed. AI recruitment and data-pipeline development.

AI Counseling Chatbot for Schizophrenia Care. Jeonbuk National University Hospital | Ongoing. Develop and study a free patient-facing LLM/RAG chatbot, focusing on perceived judgment, stigma-related engagement barriers, disclosure, and supportive interaction design.

Smart Mobility Safety Services and Infrastructure for Autonomous Driving. MOTIE / KIAT, International Industrial Technology Cooperation | Project P0025238. 2025 | Technical development and patent contribution. Developed OSM-supported road-detection and road-geometry algorithms and physics/XGBoost-based speed recommendations with KAIST and industry collaborators.

Public Bike-Sharing and Transit Integration for Local Mobility and Consumer Activity. Yuseong District Office and Daejeon Transportation Corporation support | Project lead. Secured and led the project; integrated Tashu bike-sharing, transit, Shinhan Card spending, and KT population-movement data to study first/last-mile connections, local festivals, and economic activity.

Personal Mobility Device Adoption among Users and Non-users. Behavioral research | Manuscript under review. PLS-SEM and multi-group analysis of 444 respondents, including 222 users and 222 non-users; examined perceived physical risk, trust, and adoption intentions.

Multiple-RAG and Multidimensional Ontology-Based Recruitment sLLM. Didimdol R&D planning | 2026. Research and engineering design for a recruitment-specific small language model, evidence retrieval, ontology mapping, evaluation, and an end-to-end processing pipeline.

SF-Olive Smart Farm Platform for Olive Production. International Collaborative Technology Development | Planned project period Aug 2026 - Jul 2029. KAIST participating researcher role: data experimentation and bias-analysis support for a platform combining advanced sensing, blockchain, and edge AI semiconductors.

PATENTS

Ontology-Based Data Standardization for Human Resource Management. Registered. Korean patent 10-2891011-00-00 · Registered 20 November 2025. Korean application 10-2025-0100830 · Filed 24 July 2025. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

Ontology-Based Human Resource Analysis and Job Inference. Registered. Korean patent 10-3027368-00-00 · Registered 29 September 2026. Korean application 10-2025-0174900 · Filed 18 November 2025. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

LLM-Based Ontology Data Processing for Human Resource Information. Registered. Korean patent 10-3027373-00-00 · Registered 29 September 2026. Korean application 10-2025-0176365 · Filed 19 November 2025. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

Apparatus and Method for Ontology-Based Candidate Search. Registered. Korean patent 10-3035190-00-00 · Registered 13 October 2026. Korean application 10-2025-0170399 · Filed 12 November 2025. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

Resume Data Augmentation and Personalized Inference Using Relationship-Aware LLM Context. Application filed. Korean application 10-2026-0082060 · Filed 7 May 2026. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

Talent Search and Recommendation Based on Conversational Hiring-Needs Analysis. Application filed. Korean application 10-2026-0110525 · Filed 17 June 2026. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee.

Ontology-Based Data Standardization for Human Resource Management. International application filed. International application PCT/KR2026/009822 · Filed 22 July 2026. Applicant / owner: TalentSeeker Co., Ltd. · Co-inventor: Wonguk Lee. Related Korean application: 10-2025-0100830.

Road Determination Device and Method of Determining a Road. Application filed. Korean application 10-2025-0209757 · Filed 24 December 2025.

Recommended Speed Determination Apparatus and Method. Application filed. Korean application 10-2025-0207816 · Filed 23 December 2025.

Method and Apparatus for Evaluating Image Tags for AI Training. Application filed. Korean application 10-2024-0067992 · Filed 24 May 2024.

Method for Predicting Fracture of Pipe Due to Corrosion of Alloy and System Thereof. Application specification. Korean patent application specification · 17 April 2024.

RESEARCH METHODS

Methods: GIS and OSM networks; physics-based modeling and XGBoost; PLS-SEM and multi-group analysis; experiments; panel-data analysis; agent-based simulation; patent analysis. Tools: Python, pandas, data engineering, ontologies, RAG, multi-agent systems, and LLMOps.